Dear Editors,
Physical Review Letters

We submit our manuscript "*Non-Local Magic Resources for Fermionic Gaussian States*", rewritten as a Letter, for publication in Physical Review Letters. The manuscript was previously reviewed by PRX Quantum (QT10218); we enclose the reports of the second round and our point-by-point reply.

Neither report questions the significance of our results. Referee B writes that the revised manuscript "*has been substantially strengthened*" and that the new Theorem 1 and its proof address their main concern from the first round. Referee A writes that, as far as they can tell, the result proved in the Appendix is correct.

The two remaining objections concern technical details that were already present in the first version of the manuscript and were not raised in the first round. Referee A points out that, for non-contiguous bipartitions, the definition of the non-local magic contains a qubit permutation that is not Gaussian; this definition, together with our closed formula for arbitrary bipartitions, was already in the first version, where the formula was supported numerically. Referee B points out an extra factor $(1-x)^{-1/2}$ in the finite-size Jacobi weight, a typo from a change of variables which, as the Referee notes explicitly, was not raised in the first round either.

Both points have a simple solution, which leaves all the figures and applications of the manuscript unchanged. Following Referee A's own suggestion, the definitions and Theorem 1 now refer to the contiguous blocks $A = \{1, \dots, \ell\}$ and $B = \{\ell + 1, \dots, N\}$, the setting used in all the applications. The typo in the Jacobi weight has been corrected: our calculations were carried out directly on the eigenvalues of the sampled covariance matrices, and the thermodynamic-limit results are unchanged. The remaining comments of Referee A concern the presentation of the proof; we have implemented all of them, including the Referee's shorter proof of Lemma 5, made the logic of the proof, already outlined in the manuscript, more explicit, and added a statement on the use of AI tools.

In the Letter, the main results are presented in the main text; the outline of the proof, the derivation of the Page curve, the area law and further technical details are given in the End Matter, and the full proof and the remaining material in the Supplemental Material. We believe that an exact, polynomial-time formula for the non-local magic of fermionic Gaussian states, and the Page curve, critical scaling and quasiparticle picture that follow from it, will be of broad interest to the readership of Physical Review Letters.

Best regards,
Beatrice Magni, on behalf of the Authors

List of changes

- Eqs. (2)–(3) and Theorem 1: restricted to the contiguous blocks $A = \{1, \dots, \ell\}$, $B = \{\ell + 1, \dots, N\}$; argument in the End Matter ("Local Gaussian orbit") and below Eq. (S5) (Referee A, Point 1).
- Abstract, Introduction, Conclusions: "contiguous" specified accordingly (Referee A, Point 1).
- End Matter, Eqs. (11)–(13): typo in the Jacobi weight corrected (Referee B).
- SM, Sec. S1: logic of the proof made explicit; new proof of Lemma S5; Points 2–14 (Referee A).
- Minor corrections: boundary condition, time units of the quench, early-time expansion of the random circuits, caption of Fig. 3.
- Acknowledgments: statement on the use of AI tools.
- Letter format: proof outline, Page-curve derivation, area law and technical details in the End Matter; full proof, random circuits and former Appendices in the SM.

The correspondence between the equation numbers quoted in the reports and the present ones is given at the end of this reply.

Reply to Referee A

> I have two main complaints about the current state of the manuscript. The first is that I believe that the current statement of theorem 1 is explicitly false. As far as I can tell the result proved in the appendix, that Eq. (57) is the maximum of ζ_2 on the orbit (50) is correct, but this is not the same as theorem 1. I can see a few ways to fix theorem 1, but each of them restricts its scope somewhat. I have provided more details in section 2 below, and what I believe to be an explicit counterexample to the current formulation of theorem 1 in section 3.

The Referee considers the result proved in the Appendix correct. The objection concerns its identification with the definition of Eq. (5) [now Eq. (3)] for non-contiguous bipartitions, which we have removed by restricting the definitions and Theorem 1 to contiguous blocks, as the Referee suggests (Point 1).

> The other major issue is that although the manuscript lacks any AI disclosure statement, in my opinion the new appendix appears to have significant LLM input. Independently of whether the proof and exposition actually have any AI input, the parts where it "feels" AI contribute to the (already long and non-obvious) proof being very difficult to understand. I have given specific suggestions for corrections below, but overall it would be nice to have some intuition about why we are making the moves we are making, the "Odyssey" figure gives a reasonably clear overview about what we are doing, but absolutely no intuition is provided about why.

AI tools were not used substantively in the previous version of the manuscript, and the proof was written and checked by the authors; for this reason no statement was included. We have now added one in the Acknowledgments: some support from AI tools was used in the writing and presentation. The Referee's impression concerns the style of some passages, not the result of the proof, which the Referee considers correct; we have rewritten the passages indicated (Points 5, 6, 9 and 12).

Concerning the intuition, we respectfully note that the reason for the moves of the proof was already given in the manuscript. Since the optimization runs over local fermionic Gaussian unitaries, the proof is formulated in the fermionic language, and the natural candidate for the optimum is the BCS canonical form of Botero and Reznik, as stated in the Introduction ("*the minimization over local Gaussian unitaries is solved exactly by the BCS canonical form*") and below Theorem 1. All the subsequent steps serve to show that this form is the global maximizer of ζ_2 on the orbit: the roadmap of Fig. 4 [now Fig. S1] is precisely this "why". To make it more visible, we now state it explicitly in the End Matter and at the beginning of Sec. S1 of the Supplemental Material, followed by a short summary of the role of each step.

> **1.** I am fairly sure that as stated Theorem 1 is false. The problem is that the $T_{\pi_{AB}}$ in equations (5) and (7) is a non-Gaussian permutation of qubits while in the appendix

(e.g. from equation (50)) we have only Gaussian operations. The claim just under (50) "so that maximizing ζ_2 over the local Gaussian orbit is exactly the problem defined by Eq. (5)" is false sometimes. Explicitly I think (see section 3) there is a state and bipartition for which the expression in equation (5) is positive while the equation (6) from theorem 1 is zero.

An easy change which would make the theorem true is to replace qubit swap operations with fermionic-Gaussian fSwap operations but I think the authors don't want to do this because they want to use the Gaussian-restricted non-local magic as an upper bound on the general unrestricted nonlocal magic. Replacing the qubit swap operations with gaussian fSwap operations would kill this inequality because (for example) an fSwap can act non-trivially on qubits even when it acts trivially on the corresponding fermionic modes. Another change which would fix the theorem is to restrict to only considering contiguous partitions (so A consists of the first l modes and B the rest). This would somewhat limit the applicability of the theorem.

We thank the Referee for the counterexample. The identification stated below Eq. (50) requires contiguous blocks, and the counterexample uses a non-contiguous bipartition. We have adopted the second fix proposed by the Referee, which keeps the bound $M_2^{\mathrm{NL}} \leq M_2^{\mathrm{GNL}}$: the definitions, Eqs. (2)–(3), and Theorem 1 now refer to $A = \{1, \ldots, \ell\}$ and $B = \{\ell + 1, \ldots, N\}$. In the End Matter ("Local Gaussian orbit") and below Eq. (50) [now Eq. (S5)] we now show that in this case $U_{O_A} \otimes U_{O_B}$ coincides, up to a phase, with the Gaussian unitary associated with $O_A \oplus \det(O_A)\, O_B$, so that Eq. (3) is exactly the maximization over the orbit of Eq. (S5); we also note that this fails for non-contiguous bipartitions. All the applications of the manuscript use contiguous blocks, or ensembles invariant under mode permutations, so no result changes.

2. I think the middle line of equation (124) isn't exactly right, without before assuming statistical independence we can't bring the expectation value inside the product. The final line is correct, however.

We thank the Referee. Corrected [now Eq. (S75)]: the independence of the rows is used only in the last step.

3. the text under equation (60) states that the equation is just Cauchy-Binet but in fact it requires some extra reasoning on top of Cauchy-Binet (clearly, since Cauchy-Binet is about $\det(AB)$ and here we have $\det(I + AB)$), I think the easiest way is by defining block some auxiliary block matrices with twice as many rows with the extra block being an identity matrix.

Eq. (60) [now Eq. (S16)] follows from $\det(\mathbb{I} + Y) = \sum_J \det Y_{J,J}$ with $Y = X^\dagger X$, together with the Cauchy–Binet formula for each principal minor; this is now stated.

4. Lemma 1 is proved with the line "The determinantal generating function gives", which while presumably fairly standard, is sufficiently unknown in the quantum information literature to require explanation, especially since the phrase "generating function" does

not appear elsewhere in this paper, or in reference [42] which appears to be the only reference given for this part. I'm pretty sure that (59) follows from the Fourier convention written just above, but it would be nice to show the proof of this if so.

Eq. (59) [now Eq. (S15)] is now derived explicitly, from the inclusion probabilities of the projection determinantal process, $\sum_{S \supseteq T} p_S = \det K_T$ (with a reference), and the principal-minor expansion. The phrase "generating function" has been removed.

5. I would be interested in an explanation of what the phrase "The transpose, rather than the adjoint, is what remembers the particle–hole reality constraint in the fermionic application". actually means (I'm not fully convinced this means anything relevant in the context where this phrase appears). This text reads more like an LLM than anything else.

The relevant object in Lemma S2 is the transpose overlap $Q^{\mathsf{T}}Q$, the adjoint one being trivial, $Q^{\dagger}Q = \mathbb{I}$: in Proposition S2 the reality of Γ gives $Z^{\mathsf{T}}Z = \mathbb{I}$, which makes $Q^{\mathsf{T}}Q$ computable. The sentence has been replaced by this explanation.

6. In the phrase "The two chains coincide term by term, proving Eq. (108)", I am not sure what the chains are.

The two sequences of equalities for the two sides of Eq. (108). This part has been replaced, see Point 11.

7. The proof between (80) and (81) is unnecessarily convoluted. A nonstandard inner product is briefly defined and had Cauchy-Schwartz applied to it to prove that $|e_i^{\dagger} R u| \leq 1$, when we have just written that $0 \leq R \leq 1$ due to the definition of R as $1/(1 + VV^{\dagger})$ where $VV^{\dagger}$ is manifestly positive-semidefinite.

Simplified as suggested (below Eq. (S36)); the previous Eq. (81) has been removed.

8. Similarly equation (148) explicitly expands the Hilbert-Schmidt norm of a Hermitian matrix of rank 2 whose eigenvector/value decomposition we have just written down and notes that cross terms do not appear since eigenvectors are orthogonal. In my opinion equation (147) is already sufficient, (148) is unnecessary and somewhat confusing.

Removed.

9. The text "In this subsection we rewrite it exactly — no inequality is involved — as an unweighted sum of squared principal minors of a single Hermitian", is very LLM-ish and slightly confusing, (rewriting exactly would imply no inequalities are involved, I don't know why this requires emphasis)

Rephrased (beginning of Sec. S1.5).

10. Just above equation (136), the word invariant only has one v in it.

Corrected (above Eq. (S87)).

11. The proof of lemma 5 and especially Equation (115) is (subjectively, for me) borderline unreadable. I spent a fair amount of time trying to understand this proof, and I think I mostly understood what was going on, but I was not able to fully verify the proof. I suggest the authors add significantly more explanation to what is going on (e.g. what is the purpose of the various auxiliary index sets, potentially with a diagram or similar). Alternatively, while trying to understand the proof of the "left-hand side" part of the lemma, I realised its possible to use an equation similar (but not exactly the same as) to that used in the first line of equation (116) to prove both sides. This leads to a (subjectively) simpler proof, using the equation $\det(I + X) = \sum_A \det(X_{A,A})$ twice. I sketched this proof in section 4. It doesn't require any sign-averaging and has substantially fewer auxillary index sets introduced. I don't mind whether the authors use this, simplify the exposition of the current argument, or even a third argument, but the current version is not ideal.

Readability is subjective, and we find the Referee's argument no easier to follow than ours. Since the Referee represents the readership, we have nevertheless followed the suggestion and adopted the Referee's argument, after correcting it: in Eqs. (20)–(24) of the report, $\det(I - H_S H_S^\mathsf{T})$ should read $\det(I + H_S H_S^\mathsf{T})$, and the sums run over $A \subseteq S$ and $B \subseteq S^c$. The proof now consists of Eqs. (S64)–(S67); the sign-average identity previously used there is derived in Proposition S3, where it is needed.

12. I don't think the text "with no factorial normalization." in theorem 2 is necessary (at least without defining what factorial normalization would mean here).

Removed.

13. I think in equation (147) (and maybe elsewhere) the notations $\|\cdot\|_{\mathrm{HS}}$ and $\|\cdot\|_2$ are used for exactly the same thing. It would be clearer to use one consistent notation.

We now use $\|\cdot\|_{\mathrm{HS}}$ throughout.

14. The notation from equation (145) is technically consistent, but unfortunate - it would maybe make more sense to define $c_x = \sum_j x_j c_j$ rather than the current version which has a dagger on one side.

Changed to $c_x^\dagger = \sum_j x_j c_j^\dagger$ [Eq. (S96)] throughout the proof of Theorem S1.

3. Counterexample to theorem 1 as stated. I believe the following is a counterexample to theorem (1) as it is currently stated. We choose the bipartition of 4 qubits $A = \{1, 3\}$, $B = \{2, 4\}$ [...]

4. Alternative proof of Lemma 5. [...]

Used as described in our replies to Points 1 and 11.

Reply to Referee B

> The revised manuscript has been substantially strengthened. In particular, I am satisfied that the new Theorem 1 and its proof address my main concern from the first round. The revised terminology, "fermionic Gaussian-restricted non-local magic," also makes the scope of the result considerably clearer. I therefore no longer have an objection to the central theorem on the basis raised in my previous report.

We thank the Referee for their positive assessment of the revised manuscript. Following Referee A, Theorem 1 is now stated for the contiguous blocks $A = \{1, \dots, \ell\}$ and $B = \{\ell + 1, \dots, N\}$, see our reply to Referee A.

> However, while examining the revised manuscript, I identified a separate technical issue in Sec. III C concerning the finite-size distribution used for Haar-random fermionic Gaussian states. [...] Equation (11) of the revised manuscript appears inconsistent with the previous result derived by Bianchi, Hackl, and Kieburg [Phys. Rev. B **103**, L241118 (2021)] for the same Haar ensemble of pure fermionic Gaussian states. [...] Compared with Eq. (6), Eq. (11) of the manuscript therefore contains an additional factor $\prod_{m=1}^{\ell}(1 - x_m)^{-1/2}$.
>
> I would ask the authors to reconcile Eq. (11) with the known finite-size distribution of Bianchi et al.. In particular, I would appreciate clarification of the following points:
>
> 1. Is the additional factor $(1 - x_m)^{-1/2}$ in Eq. (11) a typographical or transcription error, or was this weight actually used in deriving the subsequent finite-size results?
> 2. If Eq. (11) is to be corrected, Eqs. (14)–(15), including the parameters of the Jacobi polynomials and their normalization, should be revised consistently.
> 3. The manuscript states that the finite-size expression obtained from Eq. (14) is in "perfect agreement" with the numerical data in Fig. 1(a). It would therefore be useful to specify which Jacobi weight was used in evaluating the analytical prediction and how the random Gaussian states were sampled. [...] As a direct independent check, I would also ask the authors to test at least $\ell = 1$ and $\ell = 2$ for several small system sizes $N \geq 2\ell$. [...]
> 4. The authors should reassess which subsequent Page-curve results depend on the finite-size Jacobi weight. In particular, the exact finite-N expressions should be corrected where necessary, and it should be checked explicitly whether the thermodynamic-limit results in Eqs. (16)–(18), including the quoted half-partition saturation value, remain unchanged.

We thank the Referee for spotting this typo. The additional factor was introduced in the change of variables to $x_m = \lambda_m^2$ and propagated to Eqs. (14)–(15), which we have corrected consistently with the result of Bianchi, Hackl and Kieburg, now cited [Eqs. (11)–(13), in the End Matter]. Our calculations used the eigenvalues λ_m^2 directly, so no result is affected, and the thermodynamic-limit results, Eqs. (16)–(18) [now Eq. (14)], are unchanged.

Equation numbers

The equation numbers quoted in the reports refer to the previous version; the corresponding numbers in the present version are ("S" denotes the Supplemental Material):

previous	present	previous	present
Eqs. (4), (5)	Eqs. (2), (3)	Eq. (57)	Eq. (S13)
Eqs. (6), (7)	Eq. (4), Theorem 1	Eqs. (59), (60)	Eqs. (S15), (S16)
Eq. (11)	Eq. (11)	Eqs. (80), (81)	Eq. (S36), removed
Eqs. (14), (15)	Eqs. (12), (13)	Eq. (108), Lemma 5	Eq. (S63), Lemma S5
Eqs. (16)–(18)	text above Eq. (14), Eq. (14)	Eqs. (115), (116)	replaced by Eqs. (S64)–(S67)
Eq. (50)	Eq. (S5)	Eq. (124)	Eq. (S75)
Lemma 1	Lemma S1	Eq. (136)	Eq. (S87)
Theorems 2, 3	Theorems S1, S2	Eqs. (145), (147), (148)	Eqs. (S96), (S98), removed